Slide 4: Test Strategy & Site Selection

Objective: Find the source of a power quality problem and ensure accurate measurements.

A. Systematic Measurement Methodology

1. Start close to affected equipment, then move upstream.

- Start measuring near the faulty equipment and move towards the power source.
- Identify whether the disturbance comes from the supply or the equipment.

2. Printer / UPS example.

- Transients appeared at the printer but not at the UPS output.
- The 15-foot cable was relevant to the investigation, and inline filters reduced the problem.

3. Monitor multiple points together.

- Compare readings at different locations to narrow down the source of the disturbance.
- This saves time and avoids unnecessary measurements.

B. Minimizing Measurement Errors

1. Keep leads short and straight.

- Long leads can pick up interference and affect readings.

2. Keep leads around 6 feet or less.

- According to the slide, short leads usually add no significant fast-transient error.

3. Close CT jaws fully.

- An air gap can cause errors in current magnitude and phase angle.

4. Choose the correct probe rating.

- The probe should handle the expected current and frequency.
- The slide's example is rated at 500 A RMS and 5 Hz–10 kHz.

Conclusion: Correct measurement locations and proper instrument setup give reliable results.

Slide 5: Test Duration & Safety (PPE)

Objective: Decide how long to measure and how to perform measurements safely.

A. Defining Test Duration

1. Short-term (snapshots).

- Captures the waveform and harmonics over a short period.
- Useful for steady problems but may miss intermittent faults.

2. Long-term monitoring.

- The slide recommends monitoring each point for at least one week, unless the problem is already identified.
- It helps detect problems that occur occasionally.

Examples:

- Outlet store: Three days of monitoring detected residual current, although no trip occurred.
- Motor starter: Stray ground current appeared only after one month, so a short test could miss it.

B. Safety & Field Setup Protocols

1. Use PPE.

- Wear electrical gloves, safety glasses, fire-resistant clothing, a safety helmet, and safety shoes.

2. Barricade live panels.

- Restrict access to dangerous electrical areas.

3. Have a second person present.

- They can assist in an emergency.

4. Inspect insulation and connections.

- Check insulation and keep connections tight; seal openings as required by the setup.

Conclusion: Long-term monitoring helps detect intermittent faults, while PPE and safety procedures protect personnel.

Quick revision: Remember these lines

- Slide 4: Where to measure, how to find the fault, and how to avoid measurement errors.
- Slide 5: How long to measure and how to stay safe.
- Upstream: Towards the power source.
- Transient: A sudden, short-duration electrical disturbance.
- CT: Current Transformer.
- PPE: Personal Protective Equipment.